ESOMEPRAZOLE

Before meals

IV for upper GI bleed

CHEMICAL NAME (S)-5-methoxy-2-{[(4-methoxy-3,5-dimethylpyridin-2-yl)methyl]sulfinyl}-1H-benzimidazole (magnesium)

CLASS Proton pump inhibitor (S-isomer of omeprazole)

TRADE NAMES Nexium · Esomep · Emanera · Nexiam

THERAPEUTIC USES / INDICATIONS

- GORD and erosive oesophagitis.
- *H. pylori* eradication.
- NSAID-associated ulcers (healing and prevention).
- Prevention of rebleeding after endoscopic therapy for peptic ulcer bleeding (IV); Zollinger–Ellison.

ADMINISTRATION

- **PO** at least 1 h before a meal; swallow whole or disperse pellets in water (do not crush pellets).
- **IV** injection over ≥3 min or infusion.
- Sachets for NG administration.

DOSAGES

- **Erosive oesophagitis:** 40 mg once daily for 4–8 weeks; maintenance 20 mg.
- **H. pylori:** 20 mg twice daily with antibiotics.
- **Ulcer bleeding:** 80 mg IV bolus over 30 min, then 8 mg/h infusion for 72 h, then oral.
- **Children 1–11 yr:** 10–20 mg once daily.

PHARMACOKINETICS

- **Onset:** 1–2 h. Higher, more consistent levels than omeprazole.
- Metabolised by CYP2C19 and 3A4.
- Excreted in urine. **t½:** 1–1.5 h.

PHARMACODYNAMICS

- Irreversible **H^+/K^+-ATPase** inhibition → profound acid suppression.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Headache, abdominal pain, diarrhoea.
- Nausea, flatulence.
- Dry mouth.

ADVERSE EFFECTS (SERIOUS)

- Hypomagnesaemia, B12 deficiency (long-term).
- *C. difficile*, pneumonia.
- Fractures; interstitial nephritis.

CONTRAINDICATIONS & CAUTIONS

- Hypersensitivity to PPIs.
- Use with nelfinavir.

KEY INTERACTIONS

- **Clopidogrel:** avoid (↓ effect).
- **Atazanavir, ketoconazole:** ↓ absorption.
- **Citalopram, diazepam, cilostazol:** ↑ levels.

NURSING MANAGEMENT

- IV bleeding protocol: bolus then continuous infusion; monitor Hb, BP, HR, melaena.
- Oral: give before meals.
- Review long-term need.

PATIENT EDUCATION

- Take 1 h before food.
- Report black stools or vomiting blood.
- Lifestyle measures for reflux.

KEY POINTS

- S-isomer of omeprazole.
- Avoid with clopidogrel.
- IV 80 mg + 8 mg/h after ulcer bleed.
- Long-term risks as other PPIs.

SOURCES CHECKED **[504]** Esomeprazole 40 mg powder for solution for injection/infusion (UK SmPC) **[505]** Esomeprazole 20 mg gastro-resistant capsules, hard (UK SmPC) **[506]** Gralnek et al., Endoscopy 2021

Full references p. 432–457